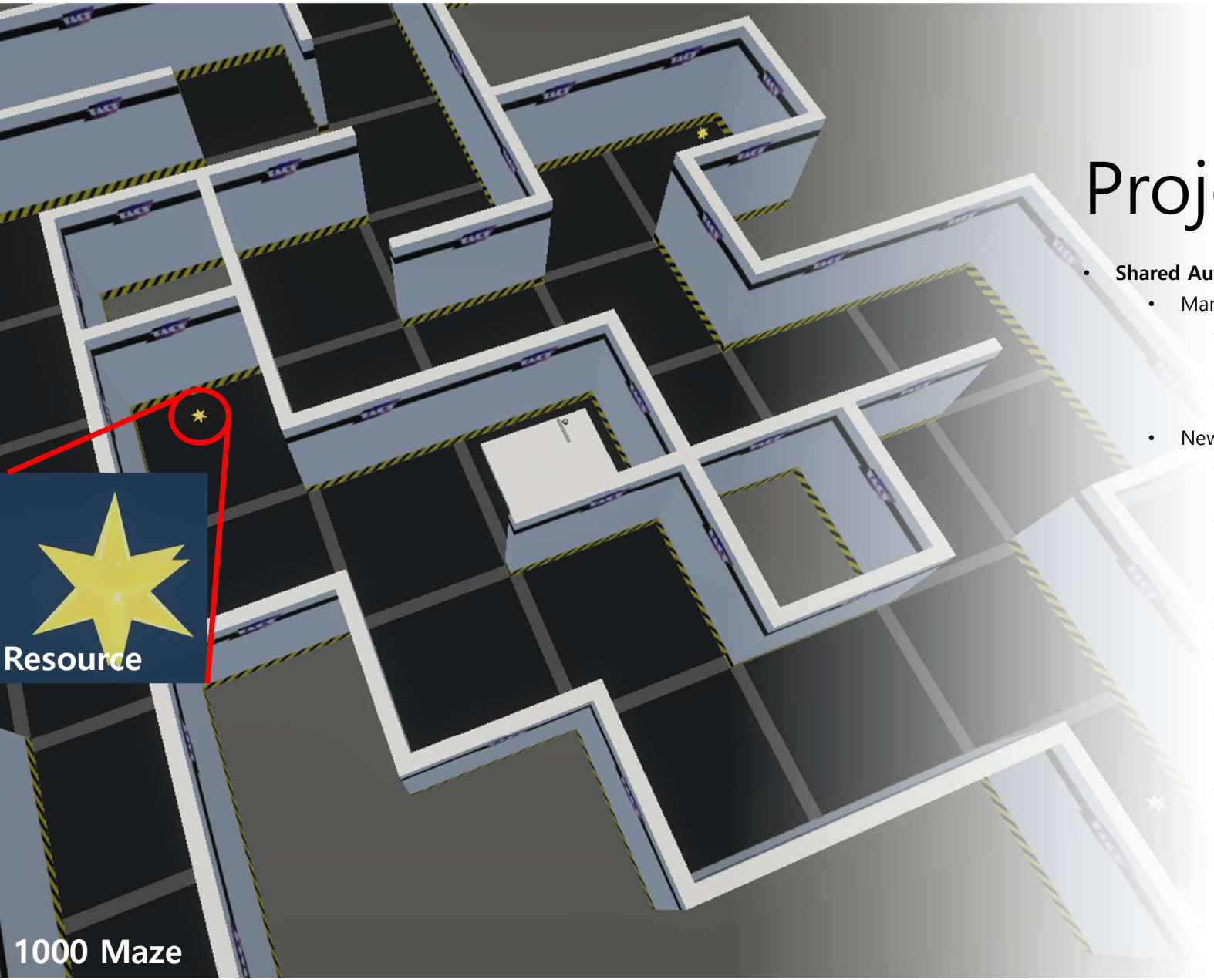
TA session 08

Project 3 – Maze



Project3

- **Shared Autonomy**
  - Manual control + Automatic control
    - Can use 1 mothership with silo, and 3 drones
    - Can use everything we've learned before
  - New "Maze" track
    - 21*21 Square grid of length 12 (wall thickness 1)
    - Random gravity and torque will be applied in each cells.
      - Use IMU.gravity, IMU.torque to use it.
    - Start form center, End in random cell
    - Must move 30 cells to reach the end
    - Always contain 7 "branches"
      - Branch does not have another branch
    - Mothership must not move at the branching cell until it checks the dead end of a branch
    - At the end of each branch, you can obtain "resource" by touching it with body or drone body.
      - It reduce your time score by 5s.
      - LIDAR.resoucePos will return the position of it when it is in range. If not, (0,0,0) will return.

New parts

- **PathPoint**
  - ControlUnit.addPathPoint(Vector3 position)
    - Place path point to certain position
  - ControlUnit.addPathPoint(Vector3 position, string message)
    - Place path point and write message on it
  - ControlUnit.removePathPoint(int ind)
    - Remove pathpoint of certain index
- **UI**
  - ControlUnit.cb.target = (drone or mothership)
    - Set camera to see target
  - ControlUnit.UI_text.text = (string)
    - Show UI text

Project3

1. **Design a mothership (Max thrust 60)**
   - Mothership must use "body" from project 1
   - Mothership can use maximum 3 drone silos (don't stack it)
2. **Design drones (Max thrust 30)**
   - Drones must use "drone body" from project 2
3. **Design manual + automatic control system**
   - You may manually control drone or automatically control mothership too
   - Consider using camera control and UI text
4. **Design controller**
   - 2 Linear potentiometer provided per team
   - **Don't forget to return every electronic parts provided for this project after presentation!**

Restrictions

If any of the restrictions are violated, the performance report will be disqualified. (0 points).

- **Overall**
  - **Mothership must not move at the branching cell until it checks the dead end of a branch.**
  - **Drone must not push mothership. Place drone low and mothership high enough so that they can not collide each other.**
  - **Do not reveal fog or use scripts to cheat the solution path of the maze.**
  - Do not drive off the given track (through a wall or across the y-axis)
  - If you want to use external libraries or assets, be sure to get prior approval.
  - You must not modify the structure, size, or components of the track or time, physics system in Unity.
- **Spacecraft**
  - Total size must fit in 9 * 9 * 5m box. **(both drone and mothership)**
  - Only one **body** part can be used in mothership, and it must be the highest parent in the hierarchy.
  - **You can only use 3 drone silo for mothership, and you must not stack them in y axis**
  - Only one **drone body** part can be used in each drone, and it must be the highest parent in the hierarchy.
  - Do not manipulate the size, weight or performance of the parts. (except frames)
  - The total thrust of all engines used in a rocket must be less than **30 for drone, 60 for mothership**.
  - A completed drone must not have any disconnected parts.
  - Except for the frame, no parts can overlap. **(servoarm will count as a frame)**
    - Frame-frame (o)
    - Frame-part (o)
    - Part-part (x)
  - Design so that interference does not occur even when the part is rotated by a servo motor.
  - **LIDAR range is 10. Do not change this value**
- **Control**
  - **You can freely control between manual and automatic for both mothership and drone. But please consider that there maybe exception you may not know.**

Evaluation criteria

- Due 12/15(Mon) 6PM
- Performance Evaluation (20)
  - Submitted Time Record (10)
    - 1000 Maze is provided. Choose **5 maps** you can perform best. Average total score will be used for the evaluation.
    - Under 10s: full score 10 (45s run, no collision, full resources)
    - Over 10s: relative evaluation
    - Cheating or violation of restriction: 0 score
  - Live Demonstration (10)
    - Your submitted control system will be demonstrated during the presentation.
    - One new map will be given for all team. (not included in given 1000 maps)
    - Criteria will be same as submitted time record.

Evaluation criteria

- Presentation Evaluation (30)
  - **Design** (8): Design of mothership, drone, control system, and controller
  - **Verification** (8): Iteration design and process
  - **Contribution** (8): Contributed part of each student
  - **Presentation** (6): Presentation and QnA

- Presentation will be held on 12/16, 18
  - Timetable TBA

Discussion Board online

- If you have any questions, please ask on blackboard discussion so that everyone in the class can see the same answer.